

BrightLight Data Analytics

SQL Exercise: Aggregate Functions & Grouping

Instructions:

1. Write on paper with a pen.
2. Draw tables of the final outcome.
3. In the **SELECT** statement, you can select the columns of your choice.
4. After completing this task, scan it to a PDF file and email it to:
rofhiwa@brightlighttutorials.co.za
5. **Submission Deadline:** Friday, March 14, 2025, 00:00

Database: employees_db

Assume you have a table called **employees** with the following structure:

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1. COUNT() Function

Write a SQL query to find the **total number of employees** in the company.

2. SUM() Function

Write a SQL query to find the **total salary** paid to all employees in the **IT department**.

3. AVG() Function

Write a SQL query to calculate the **average salary** of employees in the **HR department**.

4. MIN() and MAX() Functions

Write a SQL query to find the **highest and lowest salary** in the company.

5. GROUP BY Statement

Write a SQL query to **group employees by department** and display the **total salary paid in each department**.

6. GROUP BY and COUNT()

Write a SQL query to count how many employees work in each **city**.

7. GROUP BY and ORDER BY

Write a SQL query to group employees by **department**, calculate the **average salary** in each department, and **order the results in descending order** of average salary.

8. HAVING Clause

Write a SQL query to find **departments where the total salary paid exceeds 100,000**.
(Use GROUP BY and HAVING)

9. Combining GROUP BY, HAVING, and ORDER BY

Write a SQL query to **list cities where more than one employee works, ordered by the number of employees in descending order**.

10. Combining Aggregate Functions

Write a SQL query to find the **department with the highest average salary**.

SQL AGGREGATE FUNCTIONS & GROUPING

Q1 COUNT Function

```
SELECT COUNT(*) AS total-employees  
FROM employees;
```

Total employees
10

Q2 SUM Function

```
SELECT SUM(salary) AS total-it-salary  
FROM employees  
WHERE department = 'IT';
```

Total IT - Salary
220000

Q3 AVG Function

```
SELECT AVG(salary) AS average-hr-salary  
FROM employees  
WHERE department = 'HR';
```

Average - hr - Salary
49500

Q5 GROUP BY MIN and MAX Function

```
SELECT  
    department ,  
    SUM(salary) AS total-salary  
FROM employees  
GROUP BY department;
```

department	total - salary
Finance	119 000
HR	99 000
IT	220 000
Marketing	105 000

Q4 MIN and MAX Function

```
SELECT  
    MIN(salary) AS lowest salary,  
    MAX(salary) AS highest salary  
FROM employees;
```

lowest salary	highest salary
48 000	62 000

GROUP BY and COUNT

Q6
SELECT
city,
COUNT (*) AS employee-count
FROM employees
GROUP BY city;

City	employee-count
Chicago	3
Houston	1
Los Angeles	2
New York	2
San Francisco	2

Q7 GROUP BY and ORDER BY

SELECT
department,
AVG (salary) AS average-salary
FROM employees
GROUP BY department
ORDER BY average-salary DESC;

department	average-salary
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

Q8 HAVING CLAUSE

SELECT
department
SUM (salary) AS total-salary
FROM employees
GROUP BY department
HAVING SUM (salary) > 100 000;

department	total salary
IT	220 000
Finance	119 000
Marketing	105 000

Q9 COMBINING GROUP BY, HAVING & ORDER BY

SELECT
city,
COUNT (*) AS employees count
FROM employees
GROUP BY city
HAVING COUNT (*) > 1
ORDER BY employee-count DESC;

City	employee-count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

COMBINING AGGREGATE FUNCTIONS

Q10

```
SELECT  
  department ,  
  AVG (salary) AS highest-average-salary  
FROM employees  
GROUP BY department  
ORDER BY AVG (salary) DESC  
LIMIT 1 ;
```

department	highest-average-salary
Finance	59 500